

Chapter 4 Architecture Decision Note

Why the Cascade Controller Was Replaced by a Single-State Integrator

LLM Serving Control Project

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1 Background

Chapters 1–3 of this project applied classical cascade control to an LLM inference serving system. The inner loop regulated queue depth q by adjusting batch size B , and the outer loop regulated p95 latency ℓ by setting the queue setpoint q_{ref} . The plant model assumed:

$$\ell[k] = \alpha B[k] + \gamma B[k]^2 + \beta q[k] \quad (1)$$

with parameters $\alpha = 0.1$, $\gamma = 0.8$, $\beta = 2.0$ taken from a theoretical stochastic simulation. The key structural assumption was that a persistent request queue q builds up and directly contributes to latency via the term βq .

2 What We Found in Practice

When the cascade controller was connected to a real `qwen2.5:3b` model served by Ollama (Chapter 3), plant identification revealed two critical discrepancies.

2.1 The queue term is unidentifiable

The Stage 2 identification experiment ran sustained Poisson arrivals at $\lambda \in \{2, 3, 5, 7, 9\}$ requests per tick, with a drain rule $B[k] = \min(B_{\text{max}}, q[k] + a[k])$. In every case the measured queue depth settled to $q \approx 0$ regardless of λ .

The root cause is how Ollama processes requests internally. Unlike a production serving engine such as vLLM—which maintains an explicit scheduler queue and can hold requests pending while others are processed—Ollama dispatches each HTTP request immediately to an available inference slot. Requests that arrive when all slots are occupied are queued inside Ollama’s internal HTTP stack, not in a user-visible counter. From MATLAB’s perspective, the drain rule always matched arrivals, so the software queue counter never accumulated. The βq term in equation (1) has no observable counterpart in the real system.

2.2 TTFT is a direct function of concurrency, not queue depth

The Stage 1 identification swept concurrent batch size $B \in \{1, \dots, 12\}$ by firing B requests *simultaneously* via `parfeval`. The measured Time to First Token (TTFT) followed a clear nonlinear curve:

$$\ell(B) = c_0 + c_1 B + c_2 B^2 \quad (2)$$

with fitted parameters:

$$\begin{aligned} c_0 &= 97.16 \text{ ms} && (\text{idle prefill offset}) \\ c_1 &= 30.35 \text{ ms/req} && (\text{linear contention}) \\ c_2 &= -1.44 \text{ ms/req}^2 && (\text{saturation effect}) \end{aligned}$$

with $R^2 = 0.815$. The negative c_2 reflects the fact that at high concurrency Ollama spreads load across multiple GPU inference slots (configured via `OLLAMA_NUM_PARALLEL=16`), partially counter-acting the linear contention.

The physical mechanism is GPU KV-cache memory bandwidth competition: when B sequences are processed in parallel, each competes for bandwidth, raising TTFT. This is a concurrency effect, not a queueing effect.

2.3 The assumed β was wrong by $8\times$

Linearising equation (2) at the operating point $B_0 = 5$:

$$\beta_{\text{eff}} = \left. \frac{d\ell}{dB} \right|_{B_0} = c_1 + 2c_2 B_0 = 30.35 - 2(1.44)(5) = 15.97 \text{ ms/req} \quad (3)$$

The assumed $\beta = 2.0 \text{ ms/req}$ was $8\times$ too small. The outer-loop integral gain $K_{il} = (e^{-\Delta t/\tau} - 1)/\beta$ was therefore $8\times$ too aggressive, causing the outer integrator to over-correct on every tick.

3 Revised Architecture: Single-State Integral Controller

Since $q \approx 0$ always and the plant reduces to the static map $\ell \approx f(B)$, the cascade architecture adds no value. The inner loop was regulating a state that does not exist in the real system.

The revised Chapter 4 controller is a single-state discrete-time integrator:

$$e[k] = \ell_{\text{target}} - \ell_{\text{meas}}[k] \quad (4)$$

$$\xi[k+1] = \xi[k] + e[k] \quad (5)$$

$$B[k] = \text{clamp}(B_0 - K_{il} \xi[k], B_{\min}, B_{\max}) \quad (6)$$

The closed-loop characteristic equation is:

$$z_{\text{cl}} = 1 + \beta_{\text{eff}} K_{il} \quad (7)$$

Setting $z_{\text{cl}} = e^{-\Delta t/\tau_{\text{cl}}}$ with $\tau_{\text{cl}} = 20 \text{ s}$ and $\Delta t = 1 \text{ s}$:

$$K_{il} = \frac{e^{-\Delta t/\tau_{\text{cl}}} - 1}{\beta_{\text{eff}}} = \frac{e^{-0.05} - 1}{15.97} = -0.003054 \text{ req/ms} \quad (8)$$

The anti-windup limits on ξ are derived from the actuator bounds:

$$\xi \in \left[\frac{B_0 - B_{\max}}{K_{il}}, \frac{B_0 - B_{\min}}{K_{il}} \right] = [-1310, 2292] \quad (9)$$

4 Summary

	Chapter 3 (cascade)	Chapter 4 (single-loop)
Plant model	$\ell = \alpha B + \gamma B^2 + \beta q$	$\ell = f(B)$ (identified)
Controller states	ξ_q (inner) + ξ_l (outer)	ξ (one integrator)
Actuator	$q_{\text{ref}} \rightarrow B$ via inner loop	B directly
β used	2.0 (assumed)	15.97 (measured)
K_{il}	-0.0164 (8× too aggressive)	-0.003054 (correct)
Queue state	Tracked in software	Not needed ($q \approx 0$)

Table 1: Architecture comparison between Chapter 3 and Chapter 4.

The key insight is that **Ollama does not expose true request queueing at the application level**. TTFT is determined by GPU concurrency, not by a waiting queue. A single integral controller acting directly on batch size is both sufficient and correct for this plant. The cascade was solving a problem that did not exist in the real system.